# Number System Conversion

## 1. Converting any Base to Decimal (Base 10)

**The Concept:** This is the **Sum of Weights Method**. Every digit in a number sits in a "seat" that has a specific value (weight). To find the total value, you multiply each digit by its weight and add them all up.

### Example: Binary to Decimal

**Problem:** Convert $(1101.101)_2$ to Base 10. )

$$\begin{aligned}&(1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3}) \\ &= 8 + 4 + 0 + 1 + 0.5 + 0 + 0.125 \\ &= \mathbf{13.625_{10}}\end{aligned}$$

### Example: Hexadecimal to Decimal

**Problem:** Convert $(AF.9)_{16}$ to Base 10. *(Note: A=10,F=15)*

$$\begin{aligned}&(10 \times 16^1) + (15 \times 16^0) + (9 \times 16^{-1}) \\ &= 160 + 15 + 0.5625 \\ &= \mathbf{175.5625_{10}}\end{aligned}$$

## 2. Converting Decimal to any other Base

**The Concept:** When moving *away* from Decimal, we split the number into two separate processes:

1. **Integer Part:** Use **Repeated Division** by the target base. Record the remainders from **bottom to top**.
2. **Fractional Part:** Use **Successive Multiplication** by the target base. Record the integer parts from **top to bottom**.

### Example: Decimal to Binary (Integer Part)

**Problem:** Convert $(21)_{10}$ to Binary.

$$\begin{aligned}21 \div 2 &= 10 \text{ rem: } \mathbf{1} \text{ (LSB)} \\ 10 \div 2 &= 5 \text{ rem: } \mathbf{0} \\ 5 \div 2 &= 2 \text{ rem: } \mathbf{1} \\ 2 \div 2 &= 1 \text{ rem: } \mathbf{0} \\ 1 \div 2 &= 0 \text{ rem: } \mathbf{1} \text{ (MSB)} \\ \text{Result} &= \mathbf{10101_2}\end{aligned}$$

### Example: Decimal to Binary (Fractional Part)

**Problem:** Convert $(0.12)_{10}$ to Binary.

$$
\begin{aligned}
0.12 \times 2 &= \mathbf{0}.24 \\
0.24 \times 2 &= \mathbf{0}.48 \\
0.48 \times 2 &= \mathbf{0}.96 \\
0.96 \times 2 &= \mathbf{1}.92 \\
\text{Result} &\approx \mathbf{.0001_2}
\end{aligned}
$$

## 3. Conversions Between Arbitrary Bases

**The Concept:** If you want to convert from a "weird" base (like Base 7) to another "weird" base (like Base 20), you must use **Decimal as a bridge**. You cannot jump directly; you must go:
Base X⟶Base 10⟶Base Y

### Example: Base 7 to Base 20

**Problem:** Convert $(125)_7$ to Base 20.
**Step 1: Convert to Decimal**

$$
\begin{aligned}
&(1 \times 7^2) + (2 \times 7^1) + (5 \times 7^0) \\
&= 49 + 14 + 5 \\
&= \mathbf{68_{10}}
\end{aligned}
$$

Step 2: Convert Decimal to Base 20

$$
\begin{aligned}
68 \div 20 &= 3 \text{ rem: } \mathbf{8} \\
3 \div 20 &= 0 \text{ rem: } \mathbf{3} \\
\text{Result} &= \mathbf{38_{20}}
\end{aligned}
$$

### Example: Base 5 to Base 13

**Problem:** Convert $(324)_5$ to Base 13.
**Step 1: Convert to Decimal**

$$
\begin{aligned}
&(3 \times 5^2) + (2 \times 5^1) + (4 \times 5^0) \\
&= 75 + 10 + 4 \\
&= \mathbf{8_{10}}
\end{aligned}
$$

Step 2: Convert Decimal to Base 13

$$
\begin{aligned}
89 \div 13 &= 6 \text{ rem: } \mathbf{11} \text{ (which is )} \\
6 \div 13 &= 0 \text{ rem: } \mathbf{6} \\
\text{Result} &= \mathbf{6_{13}}
\end{aligned}
$$